Test-Time Training with Masked Autoencoders
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1. Introduction

In various data tasks, models are trained under the assumption that the data distribution encountered during training closely resembles that observed during testing. However, in real-world scenarios, this assumption is often violated due to distribution shifts. These shifts occur when the test data differs a lot from the training data, whether due to noise, corruption, etc. This issue can impact a model's ability to generalize, leading to poor performance.

One approach to addressing this issue is Test-Time Training (TTT) [4]. TTT will adapt the model during the testing phase by changing some of the model's parameter. By using test samples and auxiliary tasks, the model will adjust its representations to match the shifted distribution.

In this context, the Masked Autoencoder (MAE) [1] serves as a powerful framework for TTT. MAE is a self-supervised learning method that masks portions of input data and trains the model to reconstruct the missing parts. This reconstruction objective helps the model learn robust feature representations. When integrated with TTT, MAE enables the model to refine its features during test time, reducing the effects of distribution shifts and improving performance on test samples.

Therefore, this report examines Test-Time Training with Masked Autoencoder [2] to improve model robustness and adaptability against data corruption, noise, and distribution shifts.

2. TTT-MAE Reproduction

2.1. Experimentation setup

For the reproduction of the results presented in the research paper, we utilized the ImageNet-C dataset [3]. While the original paper [1] processed 10,000 images per corruption type, we opted to reduce this number to 100 images per corruption type. This decision was driven by the practical constraints of training time and GPU resources. Specifically, training on 1,000 images required approximately 8 to 15 hours, depending on the corruption type, making it a time-consuming and computationally expensive process. By reducing the dataset size, we prioritized conducting a broader range of experiments.

We selected four types of corruptions for our experiments: defocus blur, pixelate, brightness, and gaussian noise. To capture varying levels of difficulty, we evaluated the model on three severity levels (3, 4, and 5), where level 5 represents the most severe corruption. During test-time training, we performed 20 optimization steps per test image.

2.2. Results and analysis

To evaluate the approach, we reproduced both the TTT-MAE model and the baseline model (ViT Probing) using the parameters and configurations outlined earlier. For this, we relied on the official code provided by the research paper [1].

The results of our experiments are presented in the following graphs.

Across all types of corruptions and severity levels, TTT-MAE outperformed the baseline model. The improvement was particularly noticeable for defocus blur at level 5 and gaussian noise at level 4, where the performance gap was significant. It confirms that our reproduction aligns with the results reported in the original paper [1].

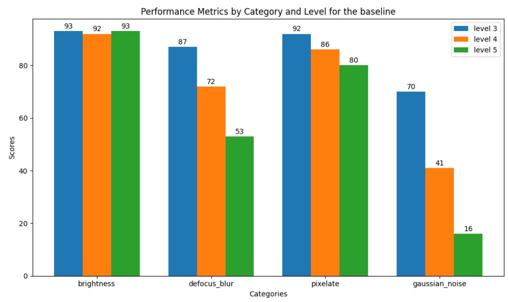
2.3. Comparison with original results

To validate our findings, we compare the results of our reproduced TTT-MAE and baseline models with the original TTT-MAE and baseline models from the research paper. The comparison focuses on the higher corruption level.

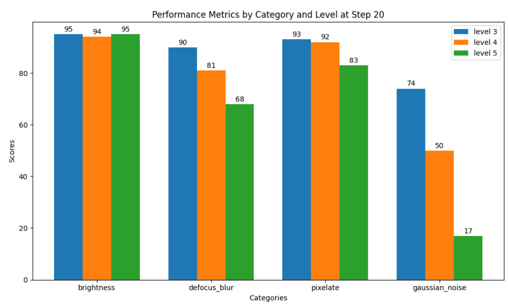
Table 1. Accuracy (%) of reproduced models vs original models for level 5.

Model	Bright.	Defocus	Pixel.	Gauss. Noise
TTT-MAE (ours)	95	68	83	17
TTT-MAE (orig.)	69.1	34.4	63	30.5
Baseline (ours)	95	53	80	16
Baseline (orig.)	68.3	24.2	49.7	17.4

Our results show that both our TTT-MAE model and



(a) Performance of the baseline across different corruption types and severity levels.



(b) Performance of TTT-MAE across different corruption types and severity levels at Step 20.

Figure 1. Reproduction results of both the baseline and the model updated using TTT-MAE framework.

baseline perform better than the original models for brightness, defocus blur, and pixelate. This improvement is due to the reduced dataset size (100 images per corruption type), which may contain simpler cases compared to the larger dataset used in the original paper. However, the trend reverses for gaussian noise, where the original TTT-MAE model outperforms our reproduction.

2.4. Impact of the steps

We analyze the impact of the number of steps during test-time training. To do this, we focused on corruption levels 5 and 4. The results are presented in two graphs, showing how the accuracy evolves as the number of steps increases for the different corruption types.

Our observations confirm that increasing the number of steps usually improves accuracy. However, the improvement is minimal or stagnates in extreme cases where the accuracy is already very high (e.g., brightness at level 5 as seen in figure 5) or very low (e.g., gaussian noise at level 5). For more regular cases, the positive effect of increasing steps is more pronounced.

These results align with the conclusions drawn in the original paper [1]. Finally, we can add that increasing the number of steps can be useful in cases where the accuracy is initially low for certain corruption types.

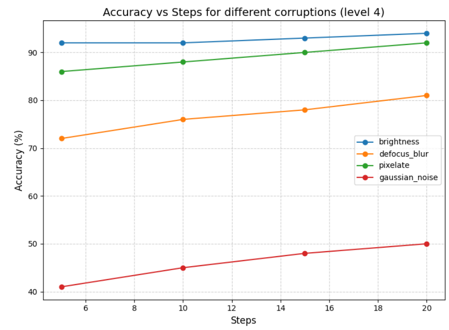


Figure 2. Impact of the number of test-time training steps on accuracy for different corruptions at severity level 4.

3. Online TTT-MAE

In the standard Test-Time Training with Masked Autoencoders (TTT-MAE) approach, the model weights are reset with each new test image, limiting the model's ability to accumulate knowledge across multiple test samples. To address this limitation, We have implemented an online adaptation variant where the model weights are retained and continuously updated throughout the test phase.

This continuous learning strategy enables the model to progressively refine its internal representations based on the sequence of test samples it encounters. By avoiding weight resets, the model can adapt more effectively to distribution shifts over time, leveraging knowledge gained from prior test samples to improve performance on subsequent inputs. This approach aligns with principles of online learning, promoting greater model robustness and adaptability in dynamic test environments.

To confirm our hypothesis, we will evaluate our models both on test accuracy but also few test time training steps accuracy over testing time.

3.1. Online Framework accuracy

Similarly to our reproduction, we tested this approach on the same subset of images and obtained better accuracies compared to the baseline and the reproduction. The results are available in the figure 3.

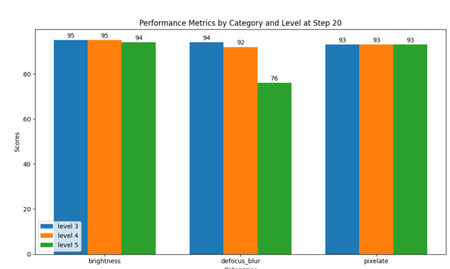
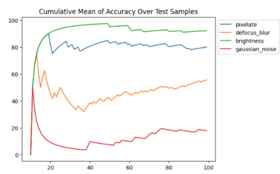


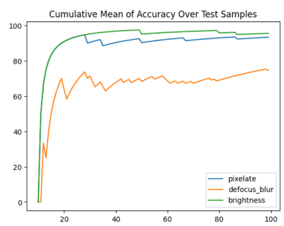
Figure 3. Performance of the online version across different corruption types and severity levels.

3.2. Few steps accuracy

One way to confirm that retaining the model weights helps to improve performance on subsequent inputs is to analyze the accuracy at 1 MAE steps over the testing dataset. We computed the cumulative average of the accuracy and we concluded that the model achieves higher accuracy and grows over testing compared to our reproduction as depicted in the figure 7.



(a) Offline



(b) Online

Figure 4. First MAE step cumulative accuracy over the testing dataset

4. Step Adaptive TTT-MAE

In the original TTT-MAE (Test-Time Training with Masked Autoencoders) framework, a fixed number of masked autoencoder (MAE) steps is applied uniformly across all test images. While this approach has shown improved generalization under distribution shifts, it does not account for the varying levels of confidence the model may have on individual test samples.

To address this limitation, I propose an adaptive strategy where the number of MAE steps performed during test-time training is dynamically adjusted based on the model's confidence score for each image. Specifically, the model's initial confidence score on a given test image can serve as a guiding metric, with lower-confidence predictions prompting additional MAE refinement steps. We re-implemented the complete data, training, and evaluation pipeline while ensuring consistency of results across experiments. For each test image, the confidence score of the most probable class inferred by the model is obtained, and the number of required steps is linearly reduced using the formula: $steps = max(1, maxSteps * (1 - score))$. This ensures performing at least one MAE step when confidence is high and the maximum number of steps when the model's confidence is low.

This formula could be further refined by considering the score to be minimal when it reflects a random choice among all classes rather than zero. By varying the number of MAE steps in accordance with the confidence score, this method introduces a more flexible and data-aware approach to test-time training, balancing computational efficiency with model reliability on challenging samples.

5. Results

In this section, we report our results over the different levels (3 to 5) and categories tested (brightness, defocus, pixelate) and compare them to the baseline model, TTT-MAE paper and our experiments.

Table 2. Accuracy (%) of reproduced models vs original models for level 5. *Accuracies of our experiment in percentage. TTT-MAE results are given on all images of the categories and with 10 steps. The Step Adaptive accuracy is given based on the last performed step.*

Level	**Baseline**	**TTT-MAE**	**Reproduction**	**Online**	**Adaptive**
Level 3	90.66	69.4	92.66	**94**	92.66
Level 4	83.33	61.66	89	**93.33**	87
Level 5	75.33	53.1	82	**93**	81

From the table 2, we can see that we successfully improved accuracy over the baseline using TTT-MAE approach and the online version outperforms the offline version. Moreover, our proposed step adaptive extension also performs the same as our reproduction for a fraction of the compute time. Additional results and experiments can be seen in the appendices.

6. Conclusion

In this project, we explored the Test-Time Training with Masked Autoencoders (TTT-MAE) framework by reimplementing the methodology presented in the original paper. Our work extended the approach in two ways. First, we developed an online version of TTT-MAE, where the model weights were not reset after each test image, enabling continuous adaptation over a sequence of test samples. Second, we proposed a dynamic test-time training strategy where the number of self-supervised adaptation steps varied based on the confidence score of the model, aiming to balance computational efficiency with model accuracy.

Our results demonstrated that the online version of TTT-MAE effectively maintained performance improvements across varying data shifts, while our adaptive strategy further optimized inference by reducing unnecessary adaptation steps on high-confidence predictions.

As our test dataset was relatively small, future work could explore further refinements in confidence-based adaptation strategies and evaluate the approach on a broader range of datasets and tasks.

References

[1] Yossi Gandelsman, Yu Sun, Xinlei Chen, and Alexei A. Efros. Test-time training with masked autoencoders, 2022. 1, 2

[2] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners, 2021. 1

[3] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *Proceedings of the International Conference on Learning Representations*, 2019. 1

[4] Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei A. Efros, and Moritz Hardt. Test-time training with self-supervision for generalization under distribution shifts, 2020. 1

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Supplementary Material

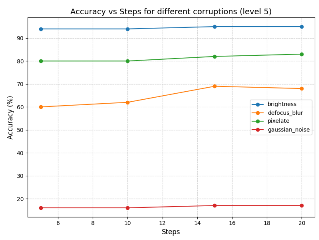


Figure 5. Impact of the number of test-time training steps on accuracy for different corruptions at severity level 5.

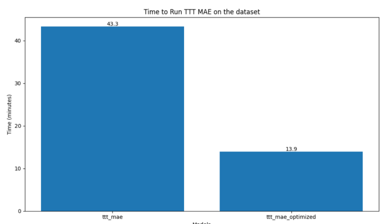
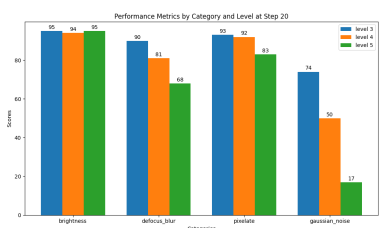
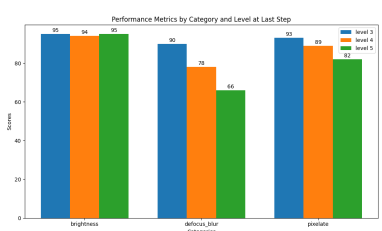


Figure 6. Test time training test time of the base version compared to our step adaptive optimized approach



(a) Reproduction



(b) Step Optimized

Figure 7. Results on different level of corruption and corruption of our reproduction compared to our optimized approach. *Accuracies for the step optimized approach are reported using the last performed MAE Step. The reproduction was ran with 20 steps.*

Figure 8. Result of the predicted image using the online, adaptative, and reproduction methods. Only the online method correctly predicted the expected image.

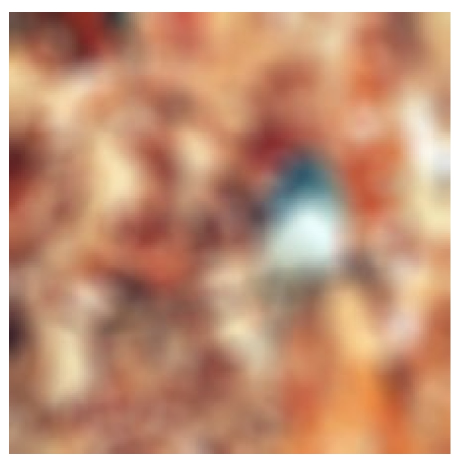


Figure 9. Result of the predicted image using the online, adaptative, and reproduction methods. Only the online method correctly predicted the expected image.